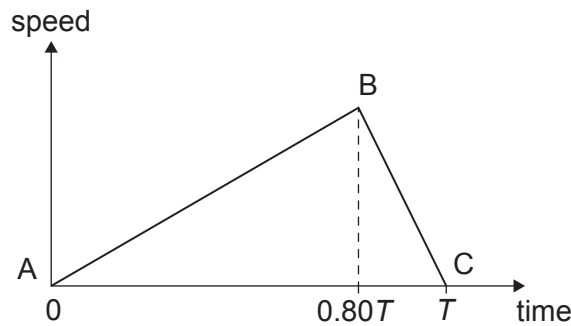


PAPER 2 / QUESTION 1

1. A train starting from rest travels from A to B with acceleration 0.20 m s^{-2} and then decelerates to rest from B to C. The total distance from A to C is 1800 m. The graph shows the variation with time of the speed of the train.

diagram not to scale



The time of travel from A to C is T and the time from A to B is $0.80 T$.

- (a) Write down the speed of the train when at point B in terms of T . [1]

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- (b) Determine T . [2]

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